

Ahmed Safierrahman Abidi

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EDUCATION

ESPRIT School of Engineering

Computer Engineering – EUR-ACE Accredited (5-year integrated diploma)

Grand Tunis, Tunisia

Sep. 2024 – Present

SAMIS High School

High School Diploma – GPA: 4.0 (99.07/100)

Saudi Arabia

Sep. 2021 – Jun. 2024

EXPERIENCE

Software Engineer Intern

Société Tunisienne de l'Electricité et du Gaz (STEG)

Jul. 2025 – Sep. 2025

Ben Arous, Tunisia

- Observed and documented the **SCADA architecture** used to supervise Tunisia's national power grid, including RTU data acquisition, IEC 60870-5 protocol framing, and real-time database management at the dispatching centre.
- Studied **Automatic Generation Control (AGC)** logic and frequency regulation mechanisms, including how the 50 Hz grid frequency is maintained within a ± 0.2 Hz band through automated correction signals to generating units.
- Examined **redundancy and failover design** across the SCADA stack: hot-standby server pairs, dual geographic dispatching centres, and RTU backup communication paths via OPGW fibre and microwave links.
- Reviewed **IT infrastructure** in a critical industrial environment, including the rationale for maintaining Red Hat Linux alongside legacy Windows XP systems, network segmentation between industrial and office environments, and cybersecurity constraints specific to operational technology.

PROJECTS

Arch Linux Mirror | GNU/Linux (Arch), rsync, systemd, NGINX, Bash, DNS, Git

Dec. 2025 – Present

- Deployed and operate the **first public Arch Linux mirror in Tunisia and North Africa**, serving 70+ GB to 1200+ unique IPs within the first two weeks with no official listing or promotion.
- Automated the full mirror lifecycle using rsync, systemd service/timer orchestration, and fail-fast execution, **reducing manual intervention to zero** across nightly upstream synchronization.
- Built a live metrics dashboard tracking requests, data served, uptime, and all-time unique IPs, accessible at mirror.safiabidi.com/dashboard.
- Navigated 6 months of ISP-level CGNAT constraints to obtain a static public IP and submitted for **official Arch Linux Tier 2 mirror listing**.

Aquarium Management System | C, PIC16F877, PCB Design, MikroC, Git

Oct. 2025 – Dec. 2025

- Built a PIC16F877-based embedded system monitoring **3 signals** (temperature, water level, lighting) and driving actuators to maintain stable conditions.
- Implemented manual override, real-time status display, and safety alarms using **internal and external interrupts**.
- Persisted thresholds and configuration across power cycles via **EEPROM**.

Starr | PHP, MySQL, HTML/CSS, JavaScript, XAMPP, Git

Oct. 2025 – Dec. 2025

- Developed a **full-stack** educational web platform supporting quizzes, messaging, and safe interaction for children in a team of 6.
- Implemented a points and badges reward system, role-based access control, and content moderation tools.

TECHNICAL SKILLS

Programming Languages: C/C++, PHP, JavaScript, HTML/CSS, SQL, Bash

Embedded: PCB Design, MikroC, MPLAB, PIC microcontrollers, interrupt systems, EEPROM

Systems & Infrastructure: Arch Linux, Debian/Ubuntu, server administration, rsync, systemd, NGINX, DNS, Git

Industrial IT Exposure: SCADA architecture, RTU systems, IEC 60870-5, AGC, network segmentation, OT cybersecurity

Languages: English (C2), Arabic (C2), French (B2)